



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
MID Exam, Trimester: Fall 2025

Course Code: CSE 113/EEE 2113; Course Title: Electrical Circuits

Total Marks: 30; Duration: 1 hour 30 min

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions.

(8 Marks)

A charge through a circuit element is shown in **Figure 1a**. The variation of power in this element is shown in **Figure 1b**.

[4+4]
CO1

- i) **Plot** the current vs time graph over the interval $0 \leq t \leq 9$ s.
- ii) **Plot** the voltage vs time graph for the same interval.

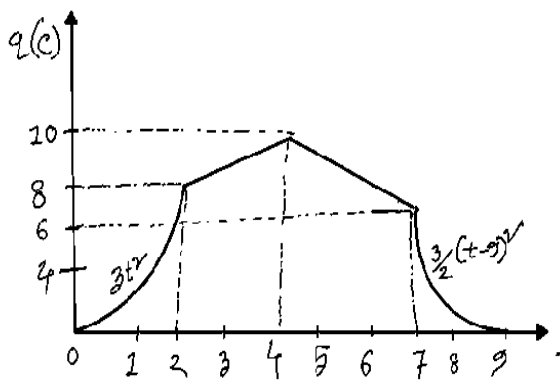


Figure 1a

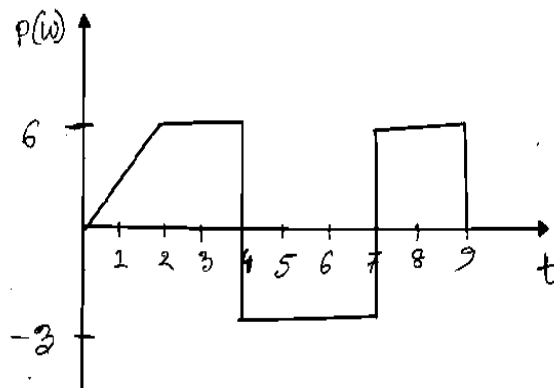


Figure 1b

Question 2: Answer all the questions.

(6 Marks)

Observe **Figure 2a** and **2b** and answer the following:

[3+3]
CO2

- i) **Apply KVL & KCL** to find the value of "**I**" in **Figure 2a**.
- ii) **Apply KVL & KCL** to find the value of "**i**" in **Figure 2b**.

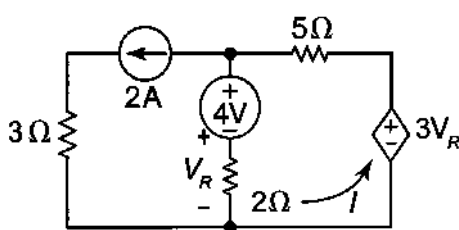


Figure 2a

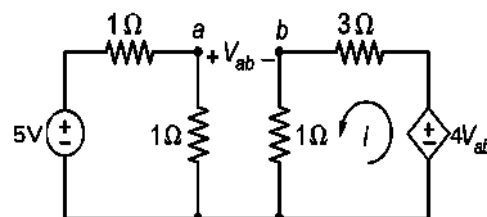


Figure 2b

Question 3: Answer all the questions

(8 Marks)

Determine the following for the circuit shown in **Figure 3**:

- The Equivalent Resistance of this circuit.
- Current i_1 and i_2 using CDR, and v_1 using VDR.

[4+4]
CO1

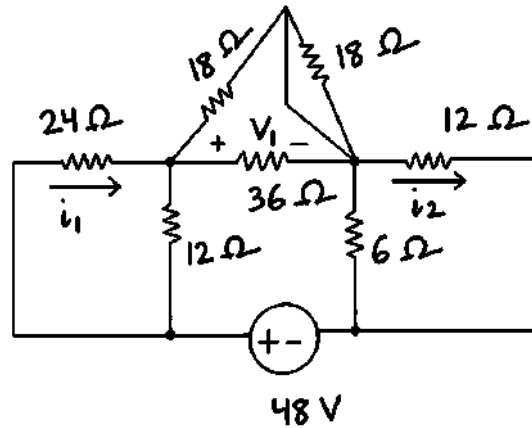


Figure 3

Question 4: Answer all the questions.

(8 Marks)

Answer the following questions for the circuit shown in **Figure 4 (a-b)**:

- Using mesh analysis in **Figure 4(a)**, find the value of V_{ab} across the dependent current source.
- Determine** the current i using nodal analysis for the circuit shown in **Figure 4(b)**.

[4+4]
CO2

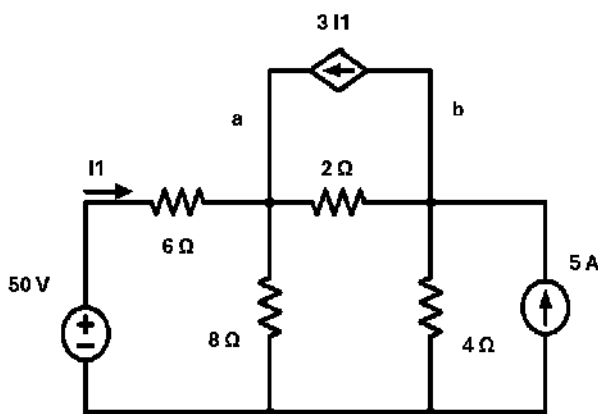


Figure 4a

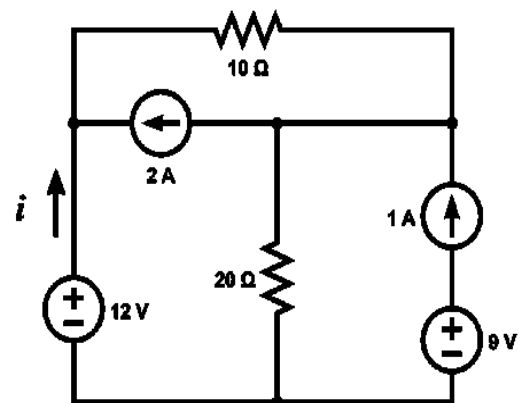


Figure 4b